

Henry Demarest

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EDUCATION

Stanford University

Palo Alto, CA

Master's Degree in Aerospace Engineering

Sept. 2025 – June, 2027

- **GPA:** 3.96
- **Awards:** 2025 National Science Foundation Graduate Research Fellowship Program (NSF GRFP)

Yale University

New Haven, CT

BS in Mechanical Engineering (ABET), BS in Computer Science

Aug. 2021 – May 2025

- **GPA:** 3.98
- **Awards:** Summa Cum Laude, Phi Beta Kappa, Tau Beta Pi, L.C. Lichty & E.O. Watters Prize in Mechanical Engineering, Matthew Isakowitz Fellowship, Robert E. Lewis Award, Wesley M. Oler Prize
- **Non-Technical Activities:** First-Year Counselor, Vice Chair of Yale ASME, MechE Peer Mentor, Engineering Tour Guide, Yale Club Running, DPods Orchestra, Backpacking Trip Leader, Intramural Secretary

WORK EXPERIENCE

Relativity Space

Long Beach, CA

GNC Engineering Intern

June 2026 - Sept 2026

- Fully owned the development, tuning, and analysis of a novel flight algorithm to be used on TerranR's ascent
- Maintained and improved a monte carlo simulation case related to pad drift on launch

GNC Simulation Engineering Intern

May 2025 - Aug 2025

- Developed an improved architecture for modeling rocket engine dynamics during flight
- Integrated an aerodynamics model for plume-body interactions during stage separation into simulation

GNC Performance Engineering Intern

May 2024 - Aug 2024

- Led key trade studies focused on optimizing vehicle trajectory via different loads-informed throttle controllers
- Implemented automated feature to improve link between monte carlo data and trajectory optimization framework

Astrobotic Technologies

Pittsburgh, PA

Lunar Surface Systems - Mechanical Engineering Intern

May 2023 - Aug 2023

- Designed test stands to subject components (solar arrays and dust covers) to cryogenic/vacuum environments
- Iterated upon the design of a spacerless-MLI thermal enclosure to enable spacecraft survival of the lunar night

PROJECTS

Yale Undergraduate Aerospace Association

New Haven, CT

Executive Board - President

May 2023 - May 2024

- Led the executive board and engaged with sponsors, administrators, aerospace companies, and other stakeholders
- Hosted inter-college conferences, industry recruitment events, volunteering opportunities, and technical workshops

Project Liquid: Chief Engineer (2023-24), Co-President (2022-23), Propulsion Lead (2021-22) *Sept 2021 - June 2025*

- Led the development of a regeneratively-cooled liquid rocket using nitrous oxide and ethane as propellants
- Managed the design, manufacturing, and testing of a custom spark ignitor
- Designed a coaxial shear injector using a stacked manifold design as Propulsion Lead

CubeSat Team – Attitude Determination and Control System (ADCS) Lead (2022-23)

Sept 2021 - June 2023

- Led the ADCS Subteam to prepare a CubeSat for launch as part of NASA's CubeSat Launch Initiative
- Designed and implemented an algorithm to control CubeSat's active stabilization efforts using magnetorquers

TECHNICAL SKILLS

Programming Languages: C, C++, D, Java, Python, MATLAB, Julia, Rust

CAD: Solidworks, Autodesk Inventor, Fusion360, OnShape

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Management (Confluence/Jira), Collaboration (GitLab, GitHub, Solidworks PDM), Microsoft Office